

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks: 25

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Prudhvi Raj.B(192324274) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Prudhvi raj.b

Annexure A

Short Answer Test

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme. (5 Marks)
2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.
 - (a) Describe the steps involved in training, validating, and evaluating the model.
 - (b) Explain how the system can assist in early warning and emergency response.(5 Marks)
3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis. (5 Marks)

Dataset:

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

4. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather, Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G)

boundaries. Show the step-by-step updates after processing each training example.

(5 Marks)

Dataset:

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

5. In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis.

(5 Marks)

Dataset:

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

6. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a person will go for a workout based on attributes such as Weather, Temperature, Time, and Mood. Apply the Find-S algorithm and determine the maximally specific hypothesis. Show the hypothesis updates after each positive example only.

(5 Marks)

Dataset:

Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes
3	Sunny	Mild	Evening	Happy	Yes
4	Cloudy	Cool	Morning	Happy	Yes

7. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples. (5 Marks) **Dataset:**

Example	Employment	Salary	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

8. In the context of the **Candidate Elimination algorithm**, consider a dataset used to classify whether a customer will subscribe to a service based on attributes such as Age Group, Income Level, Internet Usage, and Device Type. Determine the version space by computing the **specific (S)** and **general (G)** boundaries. Show step-by-step updates after each training example. (5 Marks)

Dataset:

Example	Age Group	Income Level	Internet Usage	Device Type	Subscribe
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senior	High	Occasional	Desktop	No

9. In the context of the **Candidate Elimination algorithm**, consider a dataset used to classify whether a patient follows a healthy lifestyle based on attributes such as Diet, Exercise, Sleep, and Stress Level. Apply the algorithm to compute the **version space (S and G boundaries)** and show the updates after each example. (5 Marks) **Dataset:**

Example	Diet	Exercise	Sleep	Stress Level	Healthy Lifestyle
1	Balanced	Regular	Adequate	Low	Yes
2	Balanced	Regular	Poor	Medium	Yes
3	Junk	None	Poor	High	No
4	Balanced	Moderate	Adequate	Low	Yes
5	Junk	Regular	Adequate	High	No

10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions. (5 Marks)
11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples. (5 Marks)
12. Design a machine learning model for spam email classification using labeled data. (5 Marks)
- (a) Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression).
 - (b) Describe the preprocessing steps such as tokenization, stop-word removal, and vectorization.
 - (c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score.
13. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes: (5 Marks)
- Sky, AirTemp, Humidity, Wind, Water, Forecast* and target concept *EnjoySport*. (a) Explain the initialization of S (specific boundary) and G (general boundary).
- (b) Show how S and G change with each training example.
 - (c) Compare Candidate-Elimination with the Find-S algorithm in terms of generalization capability.
14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities. (5 Marks)
- (a) Justify the choice of regression model (Linear Regression, Decision Tree, or RandomForest).
 - (b) Explain feature scaling and handling missing data.
 - (c) Describe how overfitting can be detected and avoided.
15. Explain the concept of clustering in unsupervised learning. (5 Marks)
- (a) Describe the working of the K-Means algorithm with a step-by-step example.
 - (b) Explain how the value of K is chosen using the Elbow Method.
 - (c) Compare K-Means with Hierarchical Clustering.

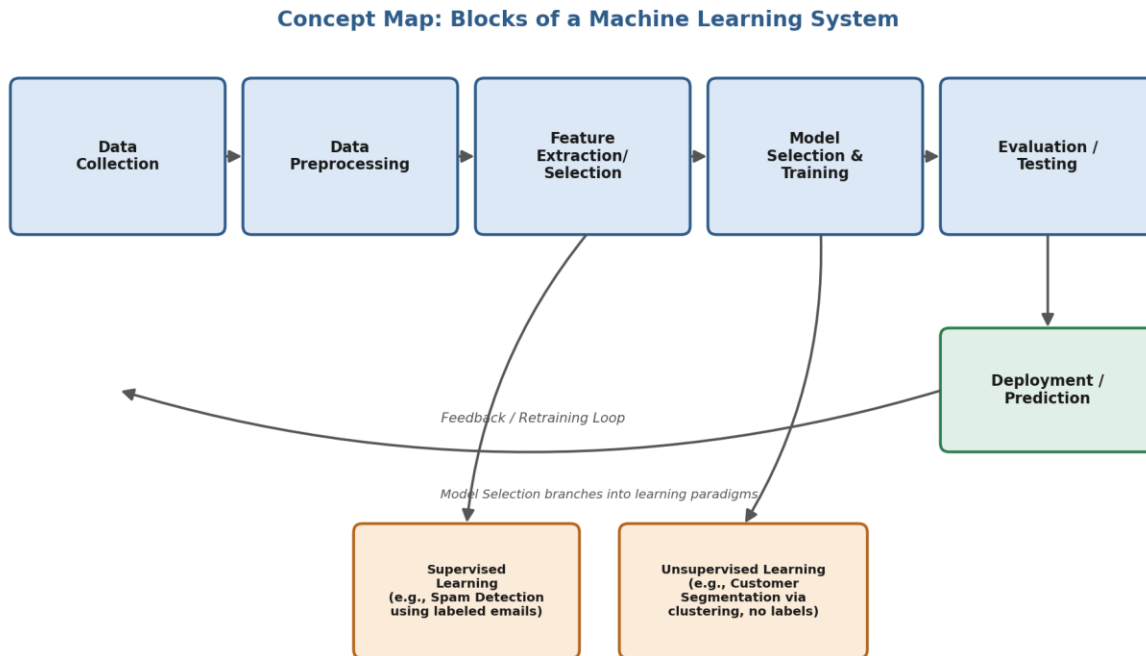
RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1 — Blocks of a Machine Learning System (5 Marks)

With the necessary sketch, explain all the possible blocks in the machine-learning system. Give examples showing the concept of supervised and unsupervised learning in the ML scheme.

Concept Map



1. Explanation of Each Block

- **Data Collection:** Gathering raw data relevant to the problem from sensors, databases, surveys, logs, or the web. Data quality here directly determines the ceiling of model performance.
- **Data Preprocessing:** Cleaning the data — handling missing values, removing duplicates/outliers, normalizing or scaling numeric features, and encoding categorical variables — so the data is consistent and usable.
- **Feature Extraction/Selection:** Identifying and selecting the most informative attributes (features) and, where needed, transforming raw data into a feature vector (e.g., converting text to TF-IDF vectors, images to pixel/embedding vectors).
- **Model Selection & Training:** Choosing an appropriate algorithm (e.g., Decision Tree, SVM, Neural Network) based on the problem type, and fitting its parameters to the training data by minimizing a loss function.
- **Evaluation/Testing:** Measuring model performance on unseen data using metrics such as Accuracy, Precision, Recall, F1-score, or RMSE, to check whether the model generalizes well.
- **Deployment/Prediction:** Integrating the trained model into a real application so it can make predictions on new, real-world input; a feedback loop often routes new data back for retraining and improvement.

2. Supervised vs Unsupervised Learning — with Examples

Supervised Learning: The model learns from labeled data, where each training example has a known input-output pair. Example: Spam Email Detection — emails labeled 'Spam' or 'Not Spam' train a classifier (e.g., Naïve Bayes) to predict labels for new emails.

Unsupervised Learning: The model works with unlabeled data and tries to discover hidden structure or patterns on its own. Example: Customer Segmentation — a retailer applies K-Means clustering on purchase history (no labels given) to group customers into segments such as 'high-spenders' and 'occasional buyers' for targeted marketing.

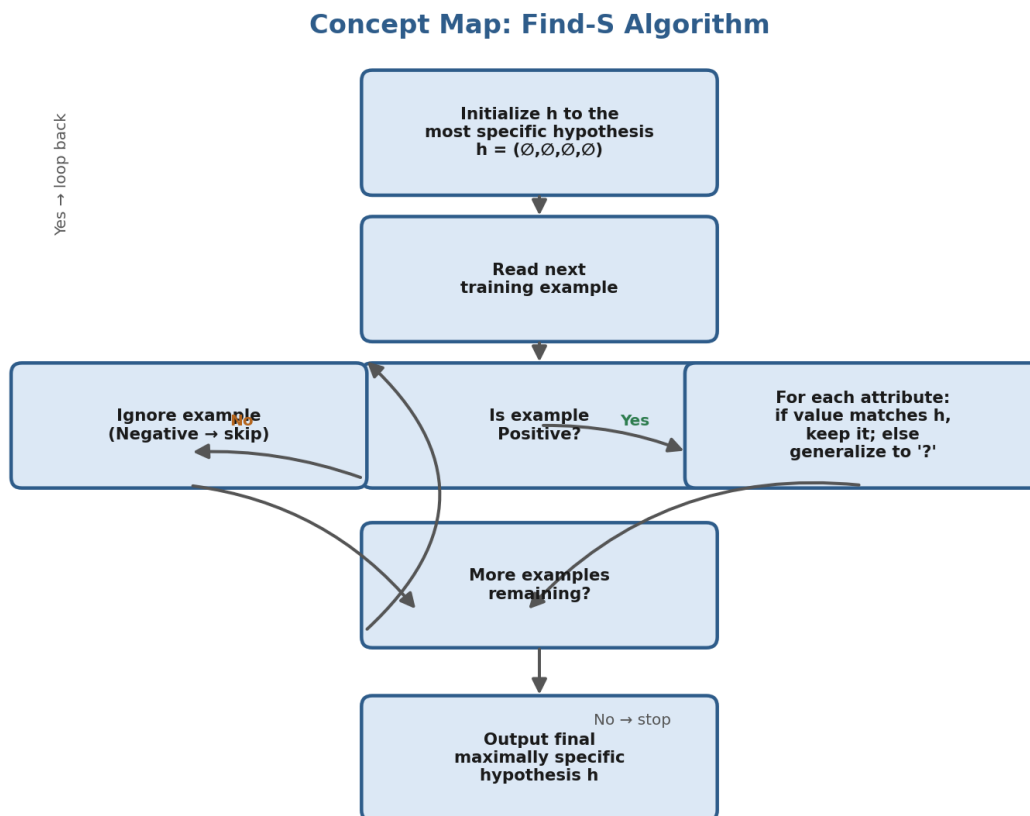
3. Conclusion

Together these blocks form an end-to-end ML pipeline: raw data is collected, cleaned, transformed into features, used to train a model matched to the learning paradigm (supervised or unsupervised), evaluated for generalization, and finally deployed — with feedback continuously improving the system.

3 — Find-S Algorithm: Play Tennis Dataset (5 Marks)

Apply the Find-S algorithm to determine the maximally specific hypothesis, showing step-by-step updates after each positive example (ignore negative examples).

Concept Map



1. Concept: How Find-S Works

Find-S starts with the most specific possible hypothesis, $h = (\emptyset, \emptyset, \emptyset, \emptyset)$, meaning 'nothing is accepted'. For every positive training example, if an attribute value in h does not match the example, that attribute is generalized to '?' (meaning 'any value is acceptable'); attributes already matching are left unchanged. All negative examples are ignored, since Find-S assumes no noise and that the target concept is expressible within the given hypothesis space.

2. Dataset

Ex	Outlook	Temp	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

3. Step-by-Step Hypothesis Updates (Positive Examples Only)

Initial hypothesis: $h_0 = (\emptyset, \emptyset, \emptyset, \emptyset)$

Example 3 (Overcast, Hot, High, Weak $\rightarrow$ Yes): h is $\emptyset$ in all fields, so it takes on Example 3's values directly.

$h_1 = (\text{Overcast}, \text{Hot}, \text{High}, \text{Weak})$

Example 4 (Rainy, Mild, High, Weak $\rightarrow$ Yes): Outlook Rainy $\neq$ Overcast $\rightarrow$ generalize to '?'. Temperature Mild $\neq$ Hot $\rightarrow$ generalize to '?'. Humidity High matches $\rightarrow$ keep. Wind Weak matches $\rightarrow$ keep.

$h_2 = (?, ?, \text{High}, \text{Weak})$

Example 5 (Rainy, Cool, Normal, Weak $\rightarrow$ Yes): Outlook already '?'. Temperature already '?'. Humidity Normal $\neq$ High $\rightarrow$ generalize to '?'. Wind Weak matches $\rightarrow$ keep.

$h_3 = (?, ?, ?, \text{Weak})$

Example 7 (Overcast, Mild, Normal, Strong $\rightarrow$ Yes): Outlook already '?'. Temperature already '?'. Humidity already '?'. Wind Strong $\neq$ Weak $\rightarrow$ generalize to '?'.
 $h_4 = (?, ?, ?, ?)$

(Examples 1, 2, 6, and 8 are negative and are ignored, as required by the algorithm.)

4. Final Hypothesis

$h_{\text{final}} = (?, ?, ?, ?)$

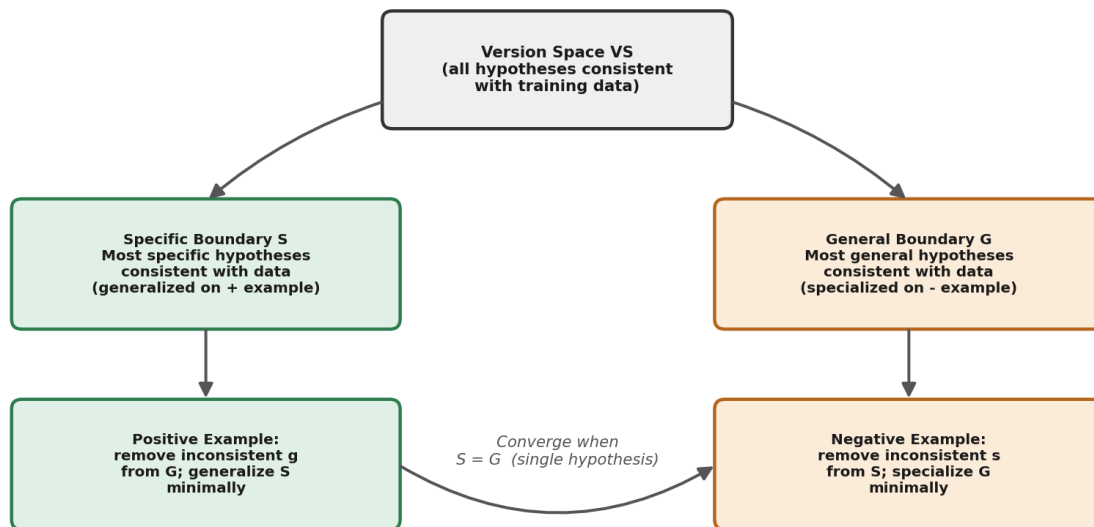
This means, based purely on the positive examples in this dataset, no attribute constrains the outcome — every combination of Outlook, Temperature, Humidity, and Wind seen among the positive examples leads to 'Play Tennis = Yes'. This signals the training data (specifically its positive examples) is too diverse for Find-S to isolate a more specific pattern; a richer dataset or the Candidate-Elimination algorithm would be needed to also use negative examples and narrow the hypothesis further.

4 — Candidate Elimination Algorithm: Enjoy Sport Dataset (5 Marks)

Using the Candidate Elimination algorithm, determine the version space by computing the specific (S) and general (G) boundaries, showing step-by-step updates after each training example.

Concept Map

Concept Map: Candidate-Elimination Algorithm (Version Space)



1. Concept: How Candidate Elimination Works

The algorithm maintains two boundaries of the version space: S (the most specific hypotheses consistent with the data so far) and G (the most general hypotheses consistent with the data). On a positive example, hypotheses in G inconsistent with it are removed, and S is generalized minimally to cover it. On a negative example, hypotheses in S inconsistent with it are removed, and G is specialized minimally to exclude it. The algorithm converges when $S = G$, or maintains a set of hypotheses (the version space) if ambiguity remains.

2. Dataset

Ex	Weather	Temp	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

3. Step-by-Step S and G Updates

Initialization: $S_0 = (\emptyset, \emptyset, \emptyset, \emptyset)$ $G_0 = (?, ?, ?, ?)$

Example 1 (Sunny, Warm, Normal, Weak $\rightarrow$ Yes, positive): S generalizes from $\emptyset$ to match the example directly. G stays maximally general (still consistent).

$S_1 = (\text{Sunny, Warm, Normal, Weak})$ $G_1 = (?, ?, ?, ?)$

Example 2 (Sunny, Warm, High, Strong $\rightarrow$ Yes, positive): Humidity (Normal $\rightarrow$ High) and Wind (Weak $\rightarrow$ Strong) mismatch S, so both attributes generalize to '?'.
 $S_2 = (\text{Sunny, Warm, ?, ?})$ $G_2 = (?, ?, ?, ?)$

Example 3 (Rainy, Cold, High, Weak $\rightarrow$ No, negative): S remains unchanged (already consistent with 'No' by not matching). G must be specialized to exclude this negative instance while still being consistent with S_2 — only attributes present in S_2 (Weather=Sunny, Temperature=Warm) can be specialized meaningfully.

$S_3 = (\text{Sunny}, \text{Warm}, ?, ?)$ $G_3 = \{(\text{Sunny}, ?, ?, ?), (?, \text{Warm}, ?, ?)\}$

Example 4 (Sunny, Cold, Normal, Weak $\rightarrow$ Yes, positive): Temperature (Warm $\rightarrow$ Cold) mismatches S, so it generalizes to '?'. In G, the hypothesis (?, Warm, ?, ?) is inconsistent with this positive example (Temperature is Cold, not Warm) and is removed.

$S_4 = (\text{Sunny}, ?, ?, ?)$ $G_4 = \{(\text{Sunny}, ?, ?, ?)\}$

Example 5 (Rainy, Warm, Normal, Strong $\rightarrow$ No, negative): $S_4 = (\text{Sunny}, ?, ?, ?)$ is already consistent (does not match Rainy), so S is unchanged. $G_4 = \{(\text{Sunny}, ?, ?, ?)\}$ is also already consistent (correctly excludes this Rainy example), so G is unchanged.

$S_5 = (\text{Sunny}, ?, ?, ?)$ $G_5 = \{(\text{Sunny}, ?, ?, ?)\}$

4. Final Version Space

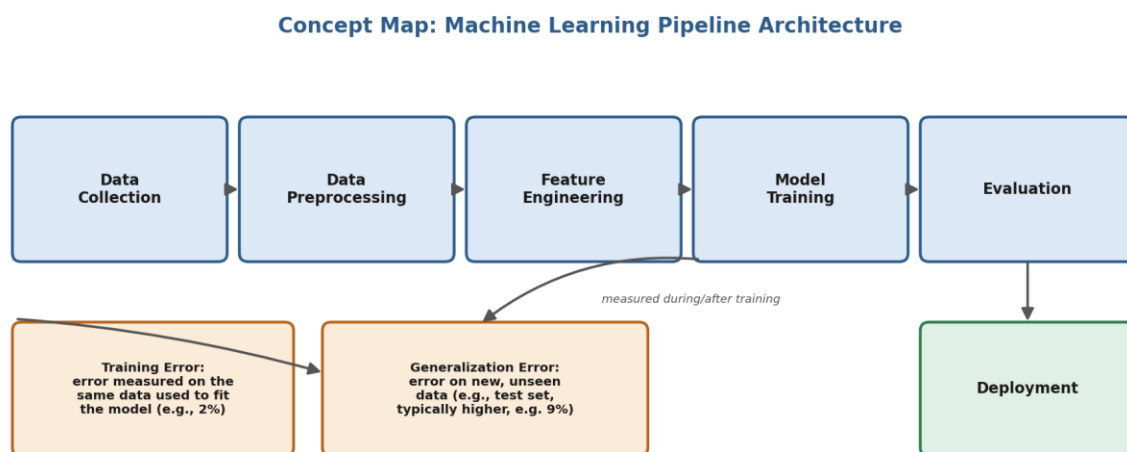
Since $S_5 = G_5 = (\text{Sunny}, ?, ?, ?)$, the algorithm has converged to a single hypothesis.

Final hypothesis: 'Enjoy Sport = Yes' if and only if Weather = Sunny, regardless of Temperature, Humidity, or Wind.

11 — Machine Learning Pipeline Architecture (5 Marks)

Explain the architecture of a typical ML pipeline with a diagram, discussing each stage, and differentiate training error from generalization error with examples.

Concept Map



1. Role of Each Pipeline Stage

- **Data Collection:** Acquiring raw data from relevant sources (databases, APIs, sensors, web scraping) that represents the problem domain.
- **Data Preprocessing:** Cleaning and transforming raw data — imputing missing values, removing noise/outliers, normalizing scales, and encoding categorical variables — into a model-ready format.
- **Feature Engineering:** Creating, transforming, or selecting the most predictive input variables (e.g., deriving 'age from date of birth', or applying PCA for dimensionality reduction) to improve model performance.

- **Model Training:** Fitting a chosen algorithm's parameters to the training data by minimizing a loss function via an optimization procedure (e.g., gradient descent).
- **Evaluation:** Assessing the trained model on a separate validation/test set using metrics like Accuracy, Precision, Recall, F1-score, or RMSE, to estimate real-world performance.
- **Deployment:** Integrating the validated model into a production system (API, application, or embedded device) so it can generate predictions on live data.

2. Training Error vs Generalization Error

Training Error is the error the model makes on the very data it was trained on. It reflects how well the model has fit the examples it has already seen. Example: a model achieves 98% accuracy on its training set.

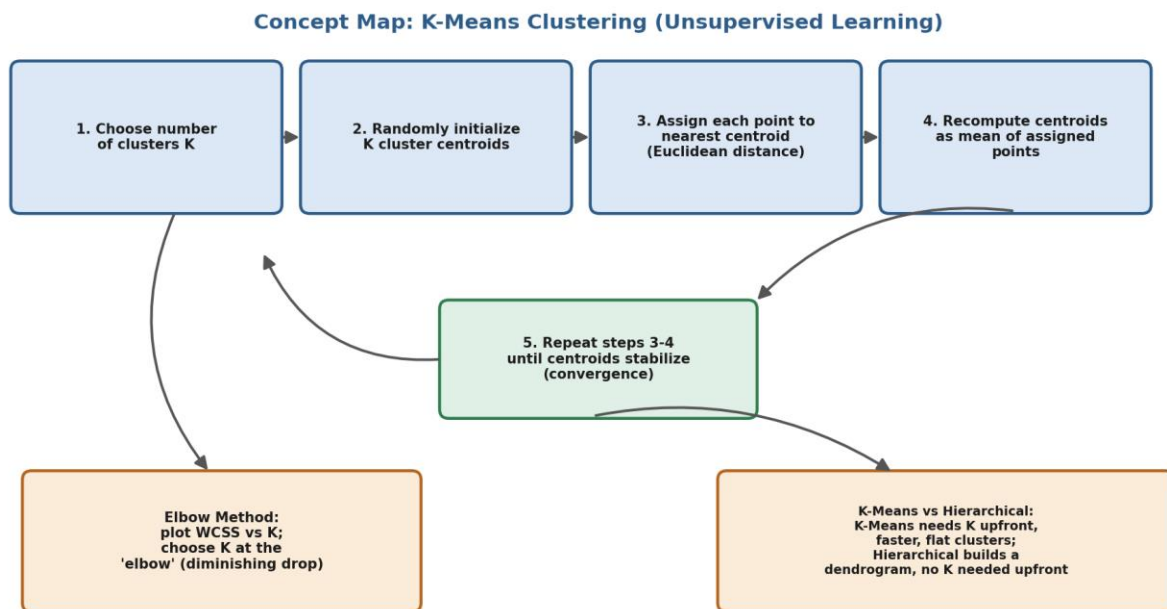
Generalization Error (or test error) is the error the model makes on new, unseen data that was not used during training. It reflects the model's real-world usefulness. Example: the same model achieves only 82% accuracy on the test set.

A large gap between the two (e.g., 98% training vs 82% test accuracy) is a classic symptom of overfitting — the model has memorized training-specific noise rather than learning generalizable patterns. A good pipeline aims to keep both errors low and close together, using techniques like cross-validation, regularization, and sufficiently diverse training data.

15 — Clustering: K-Means Algorithm (5 Marks)

Explain the concept of clustering in unsupervised learning. Describe K-Means with a step-by-step example, explain the Elbow Method for choosing K, and compare K-Means with Hierarchical Clustering.

Concept Map



1. Concept of Clustering

Clustering is an unsupervised learning technique that groups unlabeled data points into clusters such that points within a cluster are more similar to each other (by some distance/similarity measure) than to points in other

clusters. It is used for tasks like customer segmentation, image compression, and anomaly detection, where no predefined labels exist.

2. K-Means Algorithm — Step-by-Step Example

Suppose we have 6 two-dimensional points: A(1,1), B(1.5,2), C(3,4), D(5,7), E(3.5,5), F(4.5,5), and we want $K = 2$ clusters.

Step 1 — Choose K : $K = 2$.

Step 2 — Initialize centroids: randomly pick 2 points as initial centroids, e.g., $C1 = A(1,1)$ and $C2 = D(5,7)$.

Step 3 — Assign points to nearest centroid (by Euclidean distance): A, B are closer to $C1$; C, E, F, D are closer to $C2$. Cluster 1 = {A, B}, Cluster 2 = {C, D, E, F}.

Step 4 — Recompute centroids as the mean of each cluster: new $C1 = \text{mean}(A,B) = (1.25, 1.5)$; new $C2 = \text{mean}(C,D,E,F) = (4.0, 5.25)$.

Step 5 — Repeat assignment and update steps until the centroids no longer change (convergence). The final two groups represent the discovered clusters.

3. Choosing K — The Elbow Method

The Elbow Method runs K-Means for a range of K values (e.g., 1 to 10) and computes the Within-Cluster Sum of Squares (WCSS) — the total squared distance between each point and its cluster centroid — for each K . WCSS decreases as K increases, but beyond a certain point the improvement becomes marginal. Plotting WCSS against K produces a curve that bends like an elbow; the K value at this bend is chosen, since it offers a good trade-off between cluster compactness and model simplicity.

4. K-Means vs Hierarchical Clustering

Aspect	K-Means	Hierarchical Clustering
Number of clusters (K)	Must be specified in advance	Not required upfront; decided by cutting the dendrogram
Output	Flat, non-overlapping clusters	A dendrogram (tree) showing nested cluster relationships
Scalability	Efficient for large datasets ($O(n)$)	Computationally expensive for large datasets ($O(n^2)$ or more)
Sensitivity	Sensitive to initial centroid placement	Sensitive to the linkage/distance metric chosen
Reproducibility	Can converge to different results on different runs	Deterministic given the same linkage method
